
Turn in #3

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MA 542 Section A1

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Problem 1

Let $\phi : \mathbb{Q}(i) \rightarrow \mathbb{Q}$, and suppose for contradiction that $\phi \neq 0$. Then there exists $a + bi \in \mathbb{Q}(i)$ such that $\phi(a + bi) \neq 0$. Since $a, bi \in \mathbb{Q}(i)$ and ϕ is a ring homomorphism, it follows that

$$\phi(a + bi) = \phi(a) + \phi(bi) \neq 0.$$

Note that if $1 \mapsto 0$, then for all $x \in \mathbb{Q}(i)$,

$$\phi(x) = \phi(1x) = \phi(1)\phi(x) = 0\phi(x) = 0.$$

This contradicts our assumption that $\phi(a + bi) \neq 0$, so we have that $\phi(1) \neq 0$. Suppose that $\phi(1) = a$ for some $a \in \mathbb{Q}$. Note that $4 \in \mathbb{Q}$. It follows that

$$\phi(4) = \phi(1 + 1 + 1 + 1) = 4\phi(1) = 4a.$$

Additionally,

$$\phi(4) = \phi(2)\phi(2) = (2\phi(1))(2\phi(1)) = 4a^2.$$

Therefore, $a^2 = a$, and since $a \in \mathbb{Q} \setminus \{0\}$, we have that $a = 1$. Therefore, $\phi(1) = 1$. It follows that $\phi(r) = r$ for all $r \in \mathbb{Z}$ by the same proof I gave in the previous homework. I will now generalize this proof to $\mathbb{Q}$. Note that for all $r \in \mathbb{Q}$, there exist $p, q \in \mathbb{Z}$ such that $r = p/q = pq^{-1}$, where $q \neq 0$. It follows that for all $r \in \mathbb{Q}$,

$$\phi(r) = \phi\left(\frac{p}{q}\right) = \phi(p)\phi(q^{-1}) = p\phi(q)^{-1} = pq^{-1} = \frac{p}{q}.$$

Therefore, for all $r \in \mathbb{Q}$, $\phi(r) = r$. This implies that for all $a + bi \in \mathbb{Q}(i)$,

$$\phi(a + bi) = \phi(a) + \phi(bi) = a + b\phi(i).$$

Now consider $\phi(i)$. Note that $i^2 = -1$, so

$$-1 = \phi(-1) = \phi(i^2) = \phi(i)^2 \implies \phi(i) = \sqrt{-1} = i.$$

However, $i \notin \mathbb{Q}$, which is a contradiction. Since $\phi(-1) = -1$ is guaranteed since $\phi(1) \neq 0$, and this would imply that $\phi = 0$, this contradicts our original assumption that $\phi \neq 0$, and thus ϕ is the 0 homomorphism.

Problem 2

First, the commutative case. Let $\text{char } R = 2$, and let $\phi : R \rightarrow R$ be the map $x \mapsto x^2$. It follows that for all $x, y \in R$,

$$\begin{aligned}\phi(x + y) &= (x + y)^2 \\ &= x^2 + xy + xy + y^2 \\ &= x^2 + y^2 + xy(1 + 1) \\ &= \phi(x) + \phi(y) + 0xy \\ &= \phi(x) + \phi(y);\end{aligned}$$

$$\begin{aligned}\phi(xy) &= (xy)^2 \\ &= (xy)(xy) \\ &= x^2y^2 \\ &= \phi(x)\phi(y).\end{aligned}$$

Therefore, ϕ is a ring homomorphism.

Now suppose R is noncommutative. Suppose for contradiction that ϕ is a homomorphism. It follows that

$$\begin{aligned}\phi(x + y)^2 &= (x + y)^2 \\ &= (x + y)(x + y) \\ &= x(x + y) + y(x + y) \\ &= x^2 + xy + yx + y^2 \\ &= x^2 + y^2 + xy + yx \\ &= \phi(x) + \phi(y) \\ &= x^2 + y^2.\end{aligned}$$

It follows that $xy + yx = 0$, and thus $xy = -yx$ for all $x, y \in R$.

Note once again that $\text{char } R = 2$. It follows that for any $r \in R$,

$$r + r = r(1 + 1) = 0 \implies r + r = 0 \implies r = -r.$$

Therefore,

$$xy + yx = xy - yx = 0.$$

It follows that $xy = yx$, and thus R is commutative. This contradicts the fact that R is noncommutative, and thus ϕ cannot be a homomorphism if R is noncommutative.